2023-24



DEPARTMENT OF SUSTAINABLE ENERGY ENGINEERING

INDIAN INSTITUTE OF TECHNOLOGY, KANPUR

PLACEMENT BROCHURE

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MESSAGE FROM THE HOD

On behalf of the Department of Sustainable Energy Engineering at IIT Kanpur (IITK), I welcome you to the placement session of our institute this year. I am confident that you will be employing the very best from the talent pool available in our country and IITK is a coveted place for such purpose.

The institute is celebrating its diamond jubilee this year and in its journey, it has undertaken several path-breaking educational and R&D initiatives since its inception that have been imbibed by several institutes across the country. IITK takes immense pride in providing all-rounded development to its students by blending academic and research rigor with several extra-curricular activities. The institute provides science-based engineering education to its students while inculcating human values and progressiveness in their minds.

I am sure that the second postgraduate batch of Sustainable Energy Engineering is vibrant, bright, motivated, and ready to undertake the challenges posed by the industry. The department has provided its students with rigorous training not only in theoretical and analytical skills but also a hands-on experience through unique laboratory courses. We believe our students are responsible, aware, and willing to work towards tackling the national and global energy, environment, and climate issues that we face today.

The Student Placement Office (SPO) of our institute strives to provide an interactive platform to facilitate fruitful interactions between our students and potential employers. Our emphasis is on providing an open atmosphere that can establish the connection to allow synergy between the goals, objectives, and aspirations of all the stakeholders. We fully appreciate the vital role of human resources in an organizational ecosystem and are sure that you will be able to find the most suitable candidates. We are confident that our students will prove valuable to your organization and will further the objectives of your organization.

Our endeavor has been to train our students along the excellent ethos of IITK. I am sure you will be satisfied and that they will be suited to become your future employees. I am confident that they will shine in your respective organizations, in turn, providing us the necessary strength to continue our journey of achieving excellence in human resources development.

Again, we welcome you to IIT Kanpur and wish you a pleasant experience!
Best wishes,

ASHISH GARG
Professor and Head
Sustainable Energy Engineering
IIT Kanpur

ABOUT US

For societal advancement and preserving the global and Indian citizens' quality of life, ensuring sustainable energy practices is crucial. Energy production should have a minimal environmental impact and a low carbon footprint to meet society's needs.

Developing innovative renewable energy technologies requires robust R&D infrastructure and skilled human resources via specialized academic programs. Keeping this in mind, IIT Kanpur established the pioneering Sustainable Energy Engineering department, an amalgamation of Mechanical, Electrical, Aerospace, Chemical, Material Science, and Electronics Engineering. This interdisciplinary initiative is one of its kind in India.

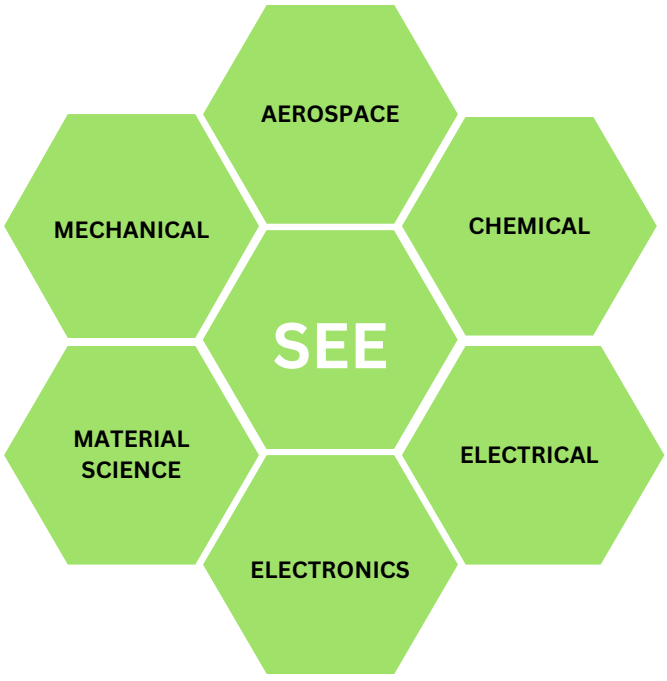
The Department of Sustainable Energy Engineering (SEE) addresses these challenges by delivering excellent education and training through dynamic postgraduate programs, fostering expertise in diverse aspects of energy sustainability.

2023-24 BATCH

16
M.Tech

07
MS(R)

10
PhD

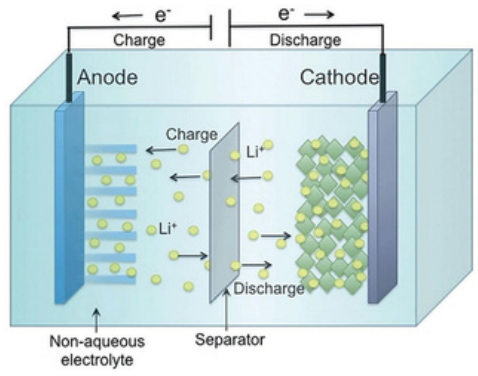


RESEARCH AREAS

- Solar - Photovoltaics & Thermal
- Hydrogen - Generation & Storage (H_2)
- Battery - Management & Manufacturing
- Wind Energy
- Green Building & Carbon Neutrality
- Energy Policy & Scenario Research
- Fuel Cells
- Power Electronics & System Integration
- Data Science & Machine Learning
- Carbon Capture and Utilization (CO_2)
- Efficient Buildings









ACADEMIC COURSES OFFERED

Our department takes immense pride in delivering industry-relevant courses that seamlessly bridge classroom learning with real-world applications. By consistently aligning our curriculum with the latest developments and demands of the ever-evolving professional landscape, we equip our students with the practical skills, up-to-date knowledge, and hands-on experiences necessary to excel in their chosen fields upon graduation. Our commitment to industry relevance ensures that students are prepared to navigate and contribute effectively to the dynamic challenges of the modern workforce.

- Introduction to Sustainable Energy Systems and Technologies (with Laboratory)
- Solar Photovoltaics
- Solar Thermal Engineering
- Hydrogen Energy
- Manufacturing of energy systems
- Smart Grid Technology
- Energy Systems: Process Analysis and Modelling
- Physics of Energy Materials
- Electrochemical Energy Systems
- Bioenergy
- Computational Methods in Engineering
- Introduction to Sustainable Energy Policy
- Energy and Carbon Markets
- Net Zero: Carbon Neutrality
- Fuel Cells
- Wind Energy
- Simulations of Power Systems
- Basics of Power Electronic Converters
- Advanced Power System Stability
- Fundamentals and Applications of Electrochemistry
- Introduction to Scientific Computing
- Essential Electrical Engineering for Renewables Integration
- Introduction to Materials Modelling and Simulations

Explore our diverse range of courses and discover how they can empower your career journey on our website [website link].



LABS / FACILITIES

TEACHING LABORATORY

The Sustainable Energy Technologies lab encompasses a range of experiments focused on instructing students in various areas such as solar photovoltaics, solar thermal techniques, energy storage, hydrogen and fuel cell technologies, smart-grid systems, wind energy, fundamental electronics, temperature, and flow gauging, as well as materials creation and analysis. This lab is part of an interdisciplinary department that has recently been set up. It offers students the opportunity to utilize a variety of laboratories and resources available at IIT Kanpur, all based on ongoing research projects.

KEY RESEARCH LABORATORIES

- Solar Photovoltaics Fabrication Laboratory
- Photovoltaics, Ultra-fast LEDs, and Storage of Energy Research Laboratory (PULSE)
- Advanced Combustion and Acoustic Laboratory (ACAL)
- Solar Energy Research Enclave (SERE) Laboratory
- Battery materials and cell development and characterization laboratory
- Hydrogen Generation and storage laboratory
- Thermal Systems Laboratory (TheSLa)
- Smartgrid Facilities
- Solar Thermal Storage
- Wind Tunnel Facilities









COLLABORATIONS

CHANDRAKANTA KESAVAN CENTER FOR ENERGY POLICY AND CLIMATE SOLUTIONS

The Center spearheads IIT Kanpur's commitment to being carbon neutral. This allows IIT Kanpur to be a showcase testbed for the transition to a low and eventually zero-carbon campus. This entails studies related to in-campus electricity consumption, conservation and management (primarily for lighting and cooling), mobility or transportation, water, and waste recycling, followed by policy intervention and implementation. This activity yields several projects that are undertaken by undergraduate and graduate students.

RICE - IITK COLLABORATIVE CENTER

Rice-IITK Collaborative Center is the first of its kind Rice, a top US university, has established a physical center with a top Indian institute. It is envisaged that the Center fosters thriving research for the development of long-term solutions to meet the ever-increasing global energy demand. Both the institutes being at the forefront of cutting-edge research in energy solutions, this Research Center facilitates deeper levels of collaboration, and exchange of knowledge and perspectives, that fertilizes ground-breaking ideas between the faculty members and researcher creating a conducive environment for both the institutes to undertake grand challenges in the area of sustainable energy and environment.



IIT KANPUR - LA TROBE UNIVERSITY RESEARCH ACADEMY



IIT Kanpur and La Trobe University have jointly set up a "Research Academy" to create a globally recognized Centre of Excellence initially focusing on Smart City Solutions.

Its mission is to foster a long-term collaboration that reflects a common research purpose which leads to collaborative research, joint training, and scholarship support for doctoral students jointly enrolled at both institutions. Working with industry and other stakeholders in India and Australia, the Research academy will facilitate the translation of research into practice and enable the commercialization of Intellectual Property.

INITIATIVES

SUSTAINABILITY CELL - IITK

Climate action and environmental sustainability are the need of the hour. And we aspire to work together for this common purpose. The primary aim of the cell is to work in close collaboration with the institute's administration and the campus residents to make the campus more sustainable and healthier to live, study & and work in. The cell also aims to:

- Foster a culture of debate & and discussions on the global polycrisis.
- Educate & and engage the community on various topics (climate change, biodiversity loss, soil degradation, consumerism, extractivism, pollution, regeneration, degrowth, etc.)

IGBC STUDENT CHAPTER

The Indian Green Building Council (IGBC), part of the Confederation of Indian Industry (CII) was formed in the year 2001. The vision of the council is, "To enable a sustainable built environment for all and facilitate India to be one of the global leaders in the sustainable built environment by 2025".

The council offers a wide array of services which include developing new green building rating programs, certification services, and green building training programs. The council also organizes the Green Building Congress, its annual flagship event on green buildings. The council is committee-based, member-driven and consensus-focused. All the stakeholders of the construction industry comprising architects, developers, product manufacturers, corporate, Government, academia, and nodal agencies participate in the council activities through local chapters.

PAST RECRUITERS



































STUDENTS' PLACEMENT OFFICE

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